Force Calculations — CIM + Koors40 + SFU1204 Ball Screw

Lead screw: **SFU1204, 4 mm lead**

CIM Motor (am-0255)		**Koors40 Controller (am-5600)**	
Rated Voltage	12 V DC	Input Voltage Range	6 V – 18 V
Free Speed	5310 rpm	Continuous Current	40 A
Free Current	2.7 A	PWM Input (normal)	1000 µs – 2000 µs
Stall Torque	2.425 N m	PWM Input (extended)	500 µs – 2500 µs
Stall Current	133 A		
Peak Power	337 W		
Shaft Diameter	8 mm (0.313 in.)		

Core Equation

$$F_{\text{piston}} \;=\; T_{\text{motor}} \cdot GR \cdot \eta_{\text{gearbox}} \cdot \frac{2\pi\,\eta_{\text{screw}}}{L} \tag{1}$$

$\eta_{\text{gearbox}} = 0.85, \quad \eta_{\text{screw}} = 0.90, \quad L = 0.004\,\text{m}$ **(SFU1204, 4 mm lead)**, $\quad \eta_{\text{combined}} = 0.765$

$$F_{\text{piston}} \;=\; T_{\text{motor}} \cdot GR \times \mathbf{1201.7} \qquad [\text{N per N m}] \tag{2}$$

Operating Point A — Koors40 continuous limit (40 A):

$$T_{\text{motor}} = k_t\,(I - I_{\text{free}}) = 0.01823 \times (40 - 2.7) = 0.680\,\text{N m}$$

Required Force at Master Cylinder

Wilwood $\frac{5}{8}$ in (15.875 mm) bore: $A_{\text{piston}} = \frac{\pi}{4}(0.015875)^2 = 1.979 \times 10^{-4}\,\text{m}^2$

Pressure	**Calc**	**Force**
300 psi (light)	2.068 MPa $\times A$	409 N
600 psi (moderate)	4.137 MPa $\times A$	819 N
1000 psi (hard)	6.895 MPa $\times A$	1365 N
1500 psi (near max)	10.342 MPa $\times A$	2047 N

Available Force @ 40 A

$F_{\text{piston}} = T_{\text{motor}} \cdot GR \times 1201.7$

GR	$F_{\textbf{piston}}$	**(lbf)**	**Pressure**	$\geq$ **2047 N**
5:1	4086 N	919	2994 psi	✓
10:1	8171 N	1837	5987 psi	✓
20:1	16 342 N	**3674**	11 975 psi	✓
50:1	40 855 N	9185	29 937 psi	✓

At the selected 20:1 ratio: $16\,342\,\text{N} \div 1.979 \times 10^{-4}\,\text{m}^2 = 11\,975\,\text{psi}$, i.e. **3674 lbf** against a **1000 lbf** operational target (**3.7× margin**) and a **5000 lbf** ball-screw axial rating.

Speed — The Cost of the Reduction

At 20:1 the screw turns at $5310/20 = 265.5\,\text{rpm}$:

$$v = \frac{265.5}{60} \times 4\,\text{mm} = 17.7\,\text{mm}\,\text{s}^{-1} \qquad \Rightarrow \qquad 25\,\text{mm stroke} = 1.41\,\text{s}$$

A shorter lead trades speed for force: versus the 5 mm assumption the actuator is **25% stronger and 20% slower**. Acceptable, because a dynamometer applies steady-state load during an RPM sweep rather than reacting to transients.

Correction Summary

Quantity	**Was (5 mm)**	**Now (4 mm)**	**Change**
Multiplier	961.3	**1201.7**	×1.25
F @ 5:1	3268 N	4086 N	×1.25
F @ 10:1	6537 N	8171 N	×1.25
F @ 20:1	13 074 N	16 342 N	×1.25
F @ 50:1	32 684 N	40 855 N	×1.25
Pressure @ 20:1	9580 psi	11 975 psi	×1.25
Screw speed @ 20:1	$22.1\,\text{mm}\,\text{s}^{-1}$	$17.7\,\text{mm}\,\text{s}^{-1}$	×0.80

L appears in the denominator, so a 20% shorter lead raises every force by exactly 25% and lowers linear speed by exactly 20%. No other term moves.